

ANALYSTLAB AFRICA

Experience Lab Internship Programme

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Week 1: FinTrust Digital Bank — Understand & Plan

Part A: Business Understanding

1. Business questions for management

- Who are FinTrust's main customer segments?
- How does transaction behaviour differ between customer segments?
- Which transaction types generate the most transaction activity and value?
- Which banking channels are most frequently used?
- What percentage of transactions are successful?
- How does transaction activity change over time?
- Which customer or transaction characteristics are associated with transactions flagged for risk review?
- Are international transactions more likely to be flagged for risk review?
- Which locations have the highest transaction activity and value?
- How does digital engagement relate to customer transaction behaviour?

2. Decisions that data analysis could support

- Customer engagement and segmentation
- Digital channel performance
- Transaction monitoring
- Operational performance
- Customer service priorities
- Identification of unusual transaction patterns
- Resource allocation
- Further investigation of potential risk indicators

3. Stakeholders who could benefit

- Senior Management
- Operations Team
- Risk Team
- Customer Experience Team
- Digital Banking Team
- Data Science Team
- Project Management Team

Part B: Data Understanding and Profiling Report

Customer Dataset

Field	Data Type	Variable Type
Customer_ID	Object/Text	Identifier
Customer_Name	Object/Text	Identifier
Age	Integer	Categorical
Gender	Object/Text	Categorical
City	Object/Text	Categorical
Customer_Segment	Object/Text	Categorical
Account_Type	Object/Text	Categorical
Tenure_Months	Integer	Numerical
Digital_Engagement_Score	Float	Numerical
Monthly_Income_Brand	Object/Text	Categorical
Preferred_Channel	Object/Text	Categorical
Account_Status	Object/Text	Categorical

Transaction Dataset

Field	Data Type	Variable Type
Transaction_ID	Object/Text	Identifier
Customer_ID	Object/Text	Foreign Key
Trasnaction_DateTime	Object/Text	Date/Time
Transaction_Type	Object/Text	Categorical
Amount_NGN	Float	Numerical
Channeel	Object/Text	Categorical
Device_Type	Object/Text	Categorical
Location	Object/Text	Categorical
International_Transaction	Object/Text	Categorical
Transaction_Status	Object/Text	Categorical
Risk_Review_Flag	Object/Text	Binary/Categorical

Missing values

There are 96 missing values in Device_Type.

There are also 96 missing values in Location.

Dataset Relationship

The two datasets share the field:

- Customer_ID

The Customer dataset contains 1,500 customers, while the Transaction dataset contains 12,000 transactions.

This represents a one-to-many relationship. This relationship allows us to combine customer information with transaction information.

Part C: Analytical Questions

1. Who are FinTrust's major customer segments?
2. How does transaction behaviour differ across customer segments?
3. Which transaction types generate the highest transaction volume and value?
4. Which channels are most frequently used and which generate the highest transaction value?
5. How does transaction activity and value change over time?
6. What proportion of transactions are successful?
7. What characteristics are associated with transactions flagged for risk review?
8. Are international transactions associated with a different risk review rate than domestic transactions?

Part D: KPI Planning

KPI	Definition	Why It Matters	Required Data
Total Transaction Value	Sum of transaction amounts	Shows the overall value flowing through the platform	Amount_NGN
Total Transactions	Count of transactions	Measures overall transaction activity	Measures overall transaction activity
Active Customers	Customers with at least one transaction	Shows customer activity	Customer_ID
Average Transaction Value	Total transaction value divided by transaction count	Shows the average transaction size	Amount_NGN, Transaction_ID
Transaction Success Rate	Successful transactions divided by total transactions $\times 100$	Measures transaction processing performance	Transaction_Status, Transaction_ID
Risk Review Rate	Transactions flagged for review divided by total transactions $\times 100$	Shows the proportion of activity requiring risk review	Risk_Review_Flag, Transaction_ID
International Transaction Rate	International transactions divided by total transactions $\times 100$	Shows the share of cross border activity	International_Transaction, Transaction_ID

Part E: Dashboard Planning

Section 1: Overall Performance

KPI cards

Total Value	Transactions	Active Customers	AVG transaction Value
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Visuals

- Transaction Value Over Time
- Transaction Volume Over Time

Section 2: Customer Intelligence

Visuals

- Customers by Segment
- Transaction Value by Customer Segment
- Transaction Activity by Account Type

Section 3: Transaction Intelligence

Visuals

- Transaction Value by Channel
- Transaction Volume by Transaction Type
- Transaction Status

Section 4: Risk Indicators

KPI cards

Risk Review Rate	International Rate
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Visuals

- Risk Review Rate by Channel
- Risk Review Rate by Transaction Type
- Risk Review Rate: International vs Domestic

Dashboard Filters

- Transaction Date
- Customer Segment
- Account Type
- Transaction Type
- Channel
- Transaction Status
- International Transaction
- Location

Major Insights

I expect the dashboard to help identify:

- Which customer segments contribute the most transaction activity and value
- Which transaction types are most common
- Which channels are most heavily used
- How transaction activity changes over time
- Whether transaction success rates vary across channels or transaction types
- Which transaction categories have higher risk review rates
- Whether international transactions show different risk review patterns
- Which locations contribute the greatest transaction activity

Initial Analysis Plan

Phase 1: Data Profiling

Profile both datasets to understand their structure, quality and relationship.

Phase 2: Data Preparation

Address confirmed data quality problems such as missing values, duplicates, invalid values and relationship issues.

Phase 3: Customer Analysis

Analyse:

- Customer segments
- Account types
- Customer locations
- Digital engagement
- Customer activity

Phase 4: Transaction Analysis

Analyse:

- Transaction volume
- Transaction value
- Transaction types
- Channels
- Transaction status
- Time trends

Phase 5: Risk Pattern Analysis

Compare risk review rates across:

- Transaction types
- Channels
- Devices
- Locations
- Transaction values
- International vs domestic transactions
- Customer segments

Phase 6: KPI Development

Create the agreed KPIs using Power BI measures.

Phase 7: Dashboard Development

Build the dashboard around the business questions rather than simply displaying every available field.

Phase 8: Business Insights

Interpret the results and translate the findings into clear business insights for management.